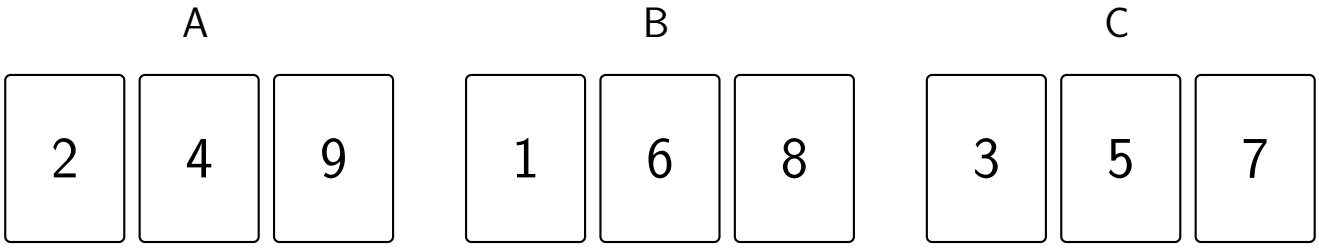
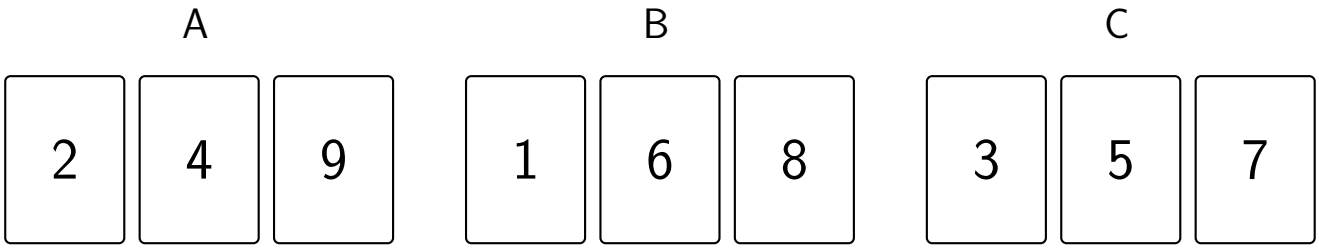
Decks A, B and C are bags of three number cards, shown below. Mix each bag on its own. In a game, each player takes one deck. Each round, draw one card from each player's bag without looking; the bigger number wins the round. Put both cards back before the next round.

Problem 1: Your partner picks one of the decks A, B, and C. Then you pick a different deck. Play ten rounds. Switch who picks first and play again. Is it better to pick first or second?

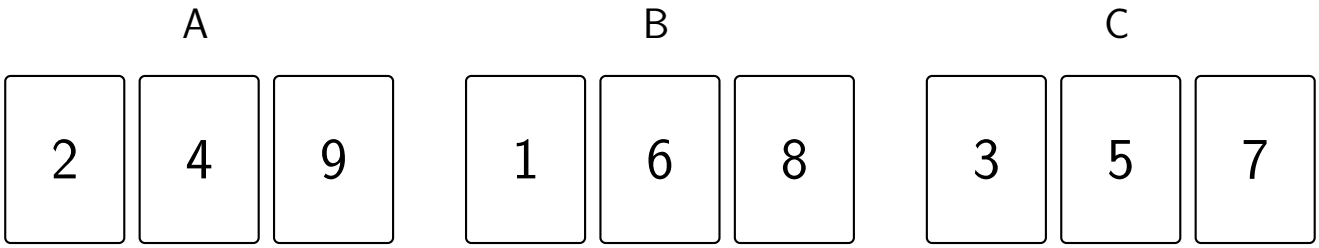


	First pick	Second pick
example	A, 4 wins	C, 6 wins

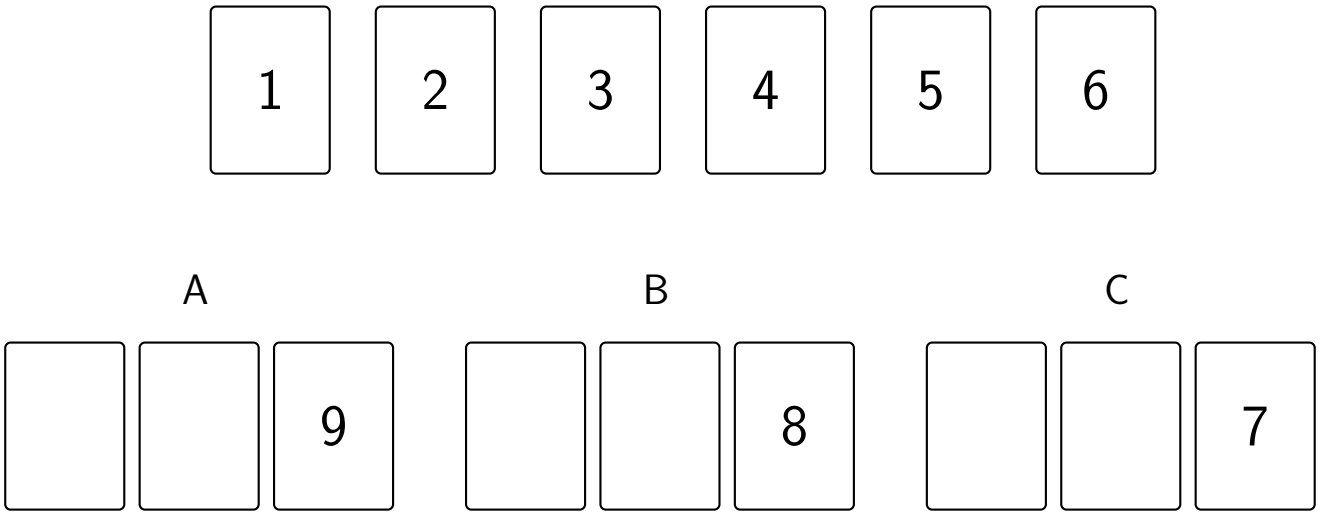
Problem 2: For each pair of decks, find which deck wins more of the possible card pairs. Is there a deck that wins more pairs than both of the others?



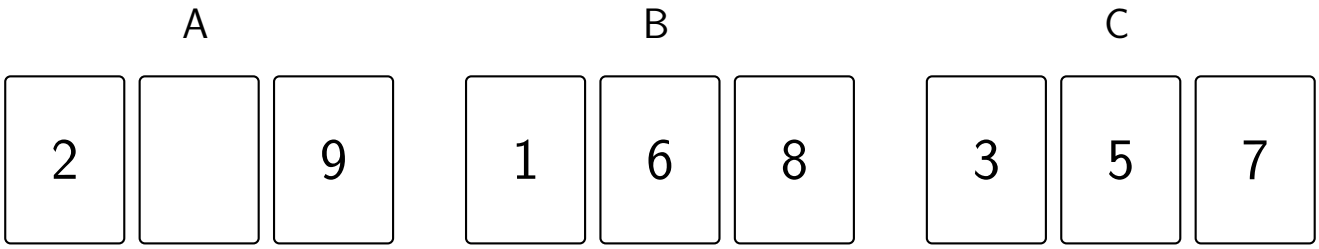
Problem 3: Your partner secretly picks two of the bags, hides their letters, and puts one on the left and one on the right. Your partner draws and tells you only which side wins. After six rounds, guess which side has the deck that wins more possible card pairs, then look. Play several times. Could any fixed number of rounds make you sure? Explain.

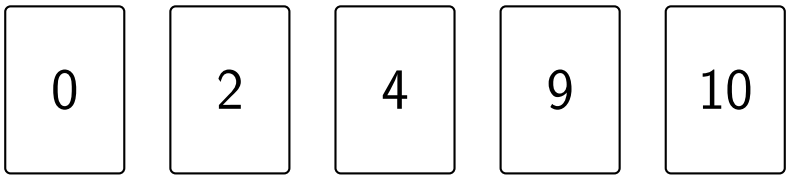


Problem 4: Keep 9 in A, 8 in B, and 7 in C. Place cards 1 to 6 so A wins more pairs than B, B more than C, and C more than A. Find two arrangements different from Problem 2.

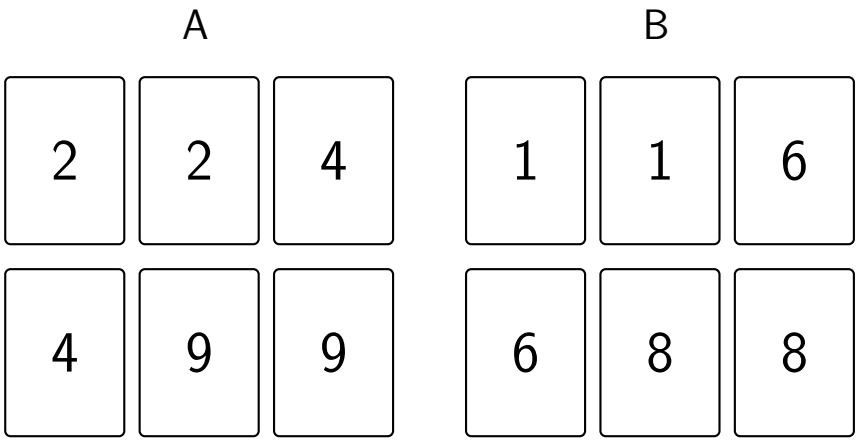


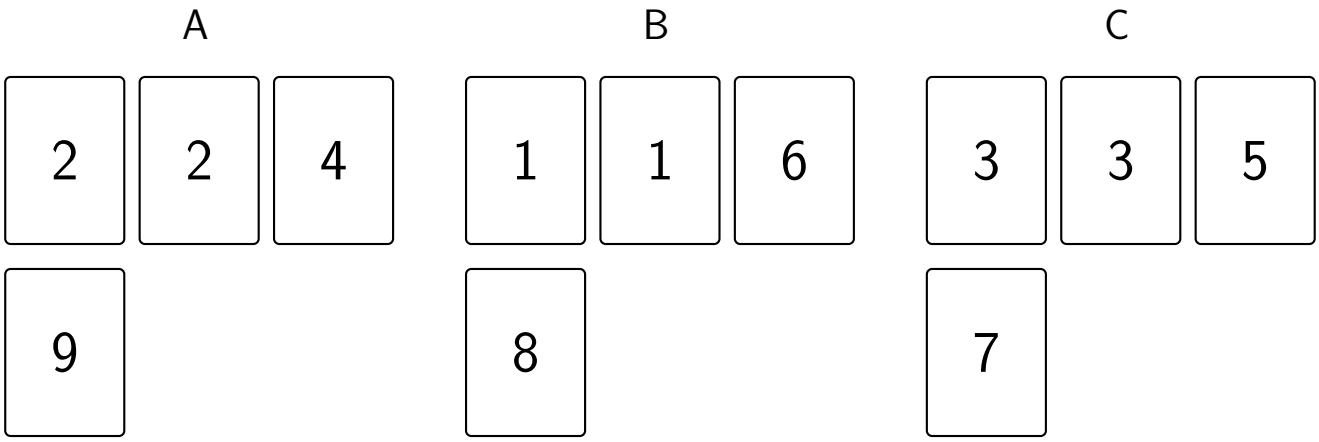
Problem 5: Choose 0, 2, 4, 9, or 10 for A's empty card. Find every choice that makes A win more pairs than B, B more than C, and C more than A.



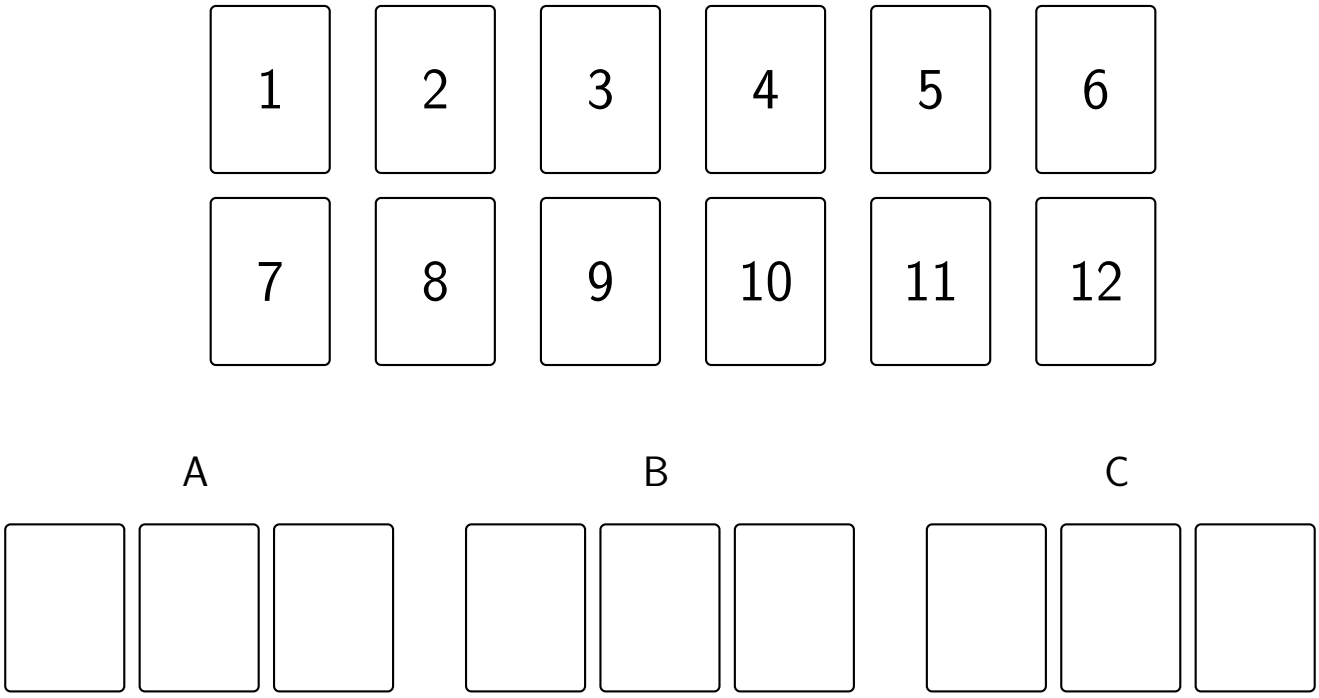


Problem 6: How many possible card pairs does A win against B with the six-card decks, and how many does B win? Then find which deck wins more pairs in each comparison of the four-card decks. Does any winner change from Problem 2?





Problem 7: Choose nine different cards from 1 to 12 and make three decks of three. Make A win more pairs than B, B more than C, and C more than A, with totals 15 in A, 18 in B, and 21 in C.



Problem 8: Use cards 1 to 6 once each to make three two-card decks. Can A win more pairs than B, B more than C, and C more than A? Find an arrangement, or explain why there is none.

